

Easy Model Comparison with Shiny Numerati Dashboard

Jo-fai (Joe) Chow

Numerai Frankfurt Community Meetup
2024-04-27



About Jo-fai (Joe) Chow

Background

Civil Engineer (Water Infrastructure)
Taught myself R & Python back in 2012
Learned machine learning from Kaggle

Work

Data Scientist / Meetup Organiser
(ex- Domino Data Lab & H2O.ai)

Contact

Twitter: [@matlabulous](https://twitter.com/matlabulous)
www.linkedin.com/in/jofaichow/



Provisional Season 2024 Status

Season Rank

9

of 514

Provisional Tier

Master

Top 10 Season Score

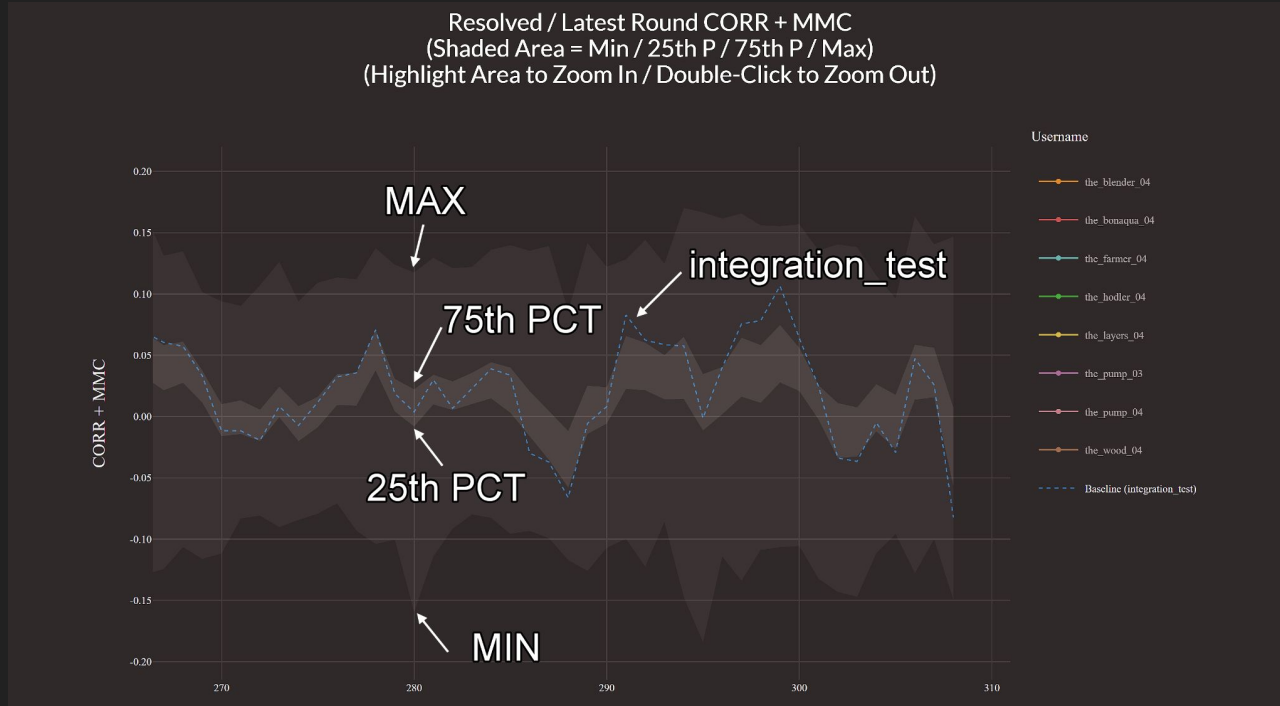
“Form is temporary, screenshot is permanent.”

About this Talk

- A Brief History of My Numerati Dashboard(s)
 - Version 1 (Static HTML)
 - Version 1 + Community Collaboration
 - Version 2 (Shiny Numerati)
- Dashboard Demo
 - KPI Comparison
 - Payout Overview

A Brief History of My Dashboards

Version One (2020 - early 2022): Numerati Dashboard



First Community Colab: Numerati + Bouwe's Streamlit

Page

tournament_view

Benchmarks

- integration_test X
- lance_a_lot X
- nekog X
- spaceshark X
- neutral_swish X

the_operator

the_pump_01

the_pump_02

the_pump_03

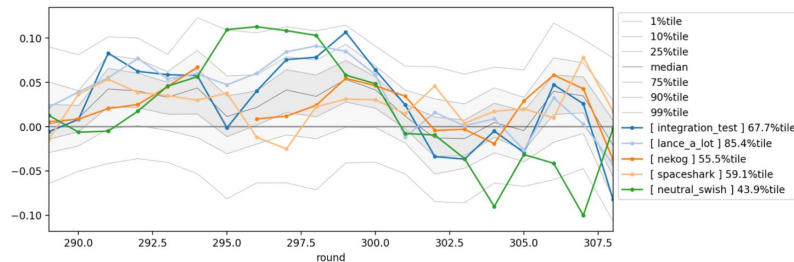
the_pump_04

the_pump_05

the_shire

the_two_towers

Tournament correlations/mmc



	COZIMMC
lance_a_lot	0.0357
integration_test	0.0269
spaceshark	0.0235
nekog	0.0220
neutral_swish	0.0171



Bouwe Ceunen

@BouweCeunen Follows you

Creator of @NumeraiPayouts. Msc. Computer Science. Writing on Medium medium.com/@bouwe.ceunen. ML/DevOps/Software Engineer.



PAYOUTS

dashboard.numeraipayouts.com

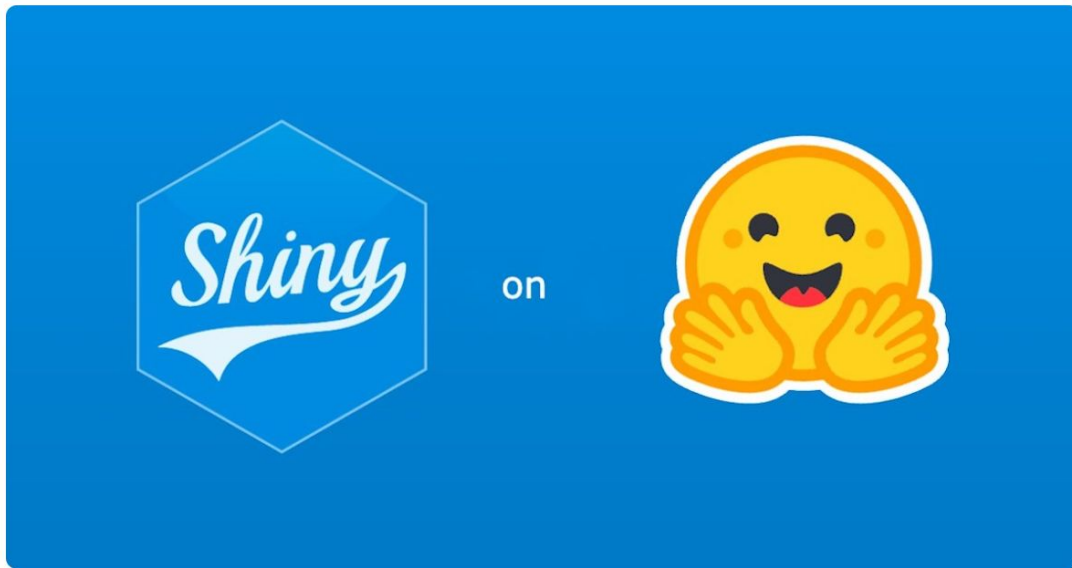
Problems with the First Version of Numerati Dashboard

- No. of models: 300 (2020) → 15000+  (Early 2022)
- Standalone HTML: 1MB → 30 MB (GitHub )
- Key metrics: MMC → TC, CORRv2
- ... so I stopped maintaining the project in early 2022

Shiny on Hugging Face

GORDON SHOTWELL

MAY 8, 2023



We're thrilled to announce that we've partnered with [Hugging Face](#) to provide templated Spaces for both the R and Python Shiny packages. These Spaces allow you to deploy an R or Python Shiny app on Hugging Face with a click of a button, and make it easy to integrate Hugging Face models into your app.

Shiny Numerati

▶ Start Here

📄 Performance Summary

📄 Raw Data

📄 Community Events

📄 About

Shiny Numerati

Community Dashboard for the Numerai Classic Tournament

Step 1: Select Your Models

First, click this ↓

120 models selected (out of 19178) ▼

Notes: 1) Use the search box below to find and select your models. 2) Use 'Select All' for quick selection. ✕

🔍 joe_the_

Select All Deselect All

joe_the_ajoe_01	✓
joe_the_ajoe_02	✓
joe_the_ajoe_03	✓
joe_the_ajoe_04	✓
joe_the_ajoe_05	✓
joe_the_blender_01	✓
joe_the_blender_02	✓
joe_the_blender_03	✓
joe_the_blender_04	✓
joe_the_blender_05	✓
joe_the_bonaqua_01	✓
joe_the_bonaqua_02	✓
joe_the_bonaqua_03	✓

Step 2: Download Data

Next, click this ↓ (it may take a while)

[Download Data from Numerai](#)



Step 1: Select Your Models

First, click this ↓

10 models selected (out of 19178)

Your Selection:

```
[1] "joe_the_force_01" "joe_the_force_02" "joe_the_force_03"  
[4] "joe_the_force_04" "joe_the_force_05" "joe_the_layers_01"  
[7] "joe_the_layers_02" "joe_the_layers_03" "joe_the_layers_04"  
[10] "joe_the_layers_05"
```

Data Preview:

CSV Excel Copy Print Show 10 entries

Search:

model	round	date_open	date_resolved	resolved	stake	corrV2	corrV2_pct	fncV3	fncV3_pct	tc	tc_pct	mmc	mmc_pct	corr_meta	apcwnm	mcwnm	pay_ft
joe_the_force_01	394	2023-01-07	2023-02-09	true	10.00	0.0261	57.6	0.0224	76.0	0.0558	90.2	0.0127	81.8	0.3966	0.2421	0.8240	0.409
joe_the_force_01	395	2023-01-10	2023-02-10	true	10.00	0.0252	77.8	0.0127	86.6	0.0997	98.6	0.0202	92.0	0.4151	0.2709	0.8219	0.000
joe_the_force_01	396	2023-01-11	2023-02-11	true	10.00	0.0225	69.3	0.0264	93.4	0.0858	96.3	0.0145	78.2	0.4230	0.2678	0.8197	0.000
joe_the_force_01	397	2023-01-12	2023-02-14	true	10.00	0.0491	96.8	0.0545	99.6	0.0298	71.1	0.0328	98.6	0.3920	0.2628	0.8236	0.000
joe_the_force_01	398	2023-01-13	2023-02-15	true	10.00	0.0457	76.3	0.0519	97.8	0.0420	83.8	0.0227	95.0	0.3970	0.2626	0.8208	0.000
joe_the_force_01	399	2023-01-14	2023-02-16	true	20.00	0.0558	90.7	0.0476	97.2	0.0826	92.7	0.0320	99.4	0.3791	0.2307	0.8222	0.411
joe_the_force_01	400	2023-01-17	2023-02-17	true	20.00	0.0412	89.1	0.0537	98.9	-0.0112	48.7	0.0273	98.1	0.4031	0.2615	0.8185	0.000
joe_the_force_01	401	2023-01-18	2023-02-18	true	20.00	0.0345	79.8	0.0576	99.5	-0.0549	20.5	0.0207	94.7	0.3944	0.2635	0.8194	0.000
joe_the_force_01	402	2023-01-19	2023-02-21	true	20.00	0.0409	78.0	0.0462	96.2	0.0018	53.4	0.0204	94.9	0.4026	0.2657	0.8229	0.000
joe_the_force_01	403	2023-01-20	2023-02-22	true	20.00	0.0306	73.1	0.0451	97.9	0.0442	87.2	0.0138	90.1	0.4006	0.2659	0.8221	0.000

Showing 1 to 10 of 3,725 entries

Previous 1 2 3 4 5 ... 373 Next

Step 2: Download Data

Next, click this ↓ (it may take a while)

Download Data from Numera



Performance Summary

Remember to refresh the charts after making changes to model selection or settings below.

NOTE: the charts may take a while to render if you have selected a lot of models.

Step 4: Adjust the Filter

Numerai Classic Tournament Rounds



Step 5: Generate Summary

 Generate / Refresh

KPI (All)

Payout (Overview)

Payout Table (Rounds)

Payout Table (Models)

Payout Chart (Rounds)

Payout Chart (Models)

Pick ONE of the KPIs:

Score Multipliers: 0.5 x CORRv2 + 2.0 x MMCv2

Cumulative

Yes

Hide Pending

No

Facet

No

Step 4: Adjust the Filter

Numerai Classic Tournament Rounds



KPI (All)

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Pick ONE of the KPIs:

Score Multipliers: 0.5 x CORRV2 + 2.0 x MMCv2

Score Multipliers: 0.5 x CORRV2 + 2.0 x MMCv2

Score Multipliers: 2.0 x CORRV2 + 1.0 x TC

MMCV2: The Latest and the Greatest MMC

Percentile: MMCv2

CORRV2: CORrelation with target cyrus_v4_20

Percentile: CORRV2

TC: True Contribution to the hedge fund's returns

Percentile: TC

FNCv3: Feature Neutral Correlation with respect to the FNCv3 features

Percentile: FNCv3

CWMM: Correlation With the Meta Model

MCWNNM: Maximum Correlation With Numerai Models staked at least 10 NMR

APCWNNM: Average Pairwise Correlation With Numerai Models staked at least 10 NMR

Payout

Rate of Return (%): Payout / Stake x 100

Cumulative

Yes

Hide Pending

No

Facet

No

KPI Data

CSV Excel Copy Print Show 5 entries

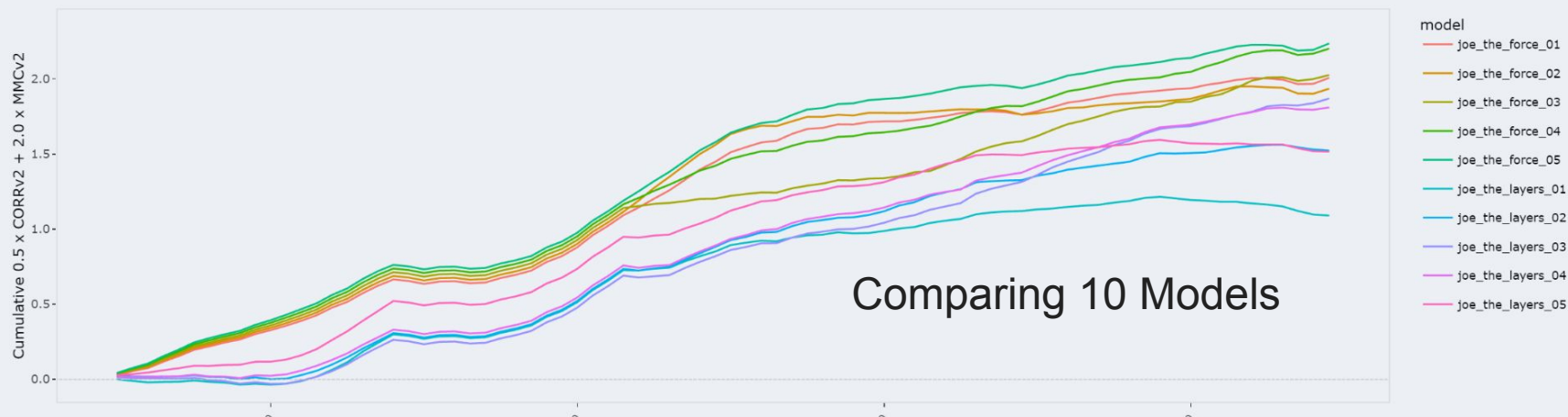
Search:

model	round	date_resolved	resolved	KPI
joe_the_force_01	650	2024-02-02	true	0.0310
joe_the_force_01	651	2024-02-03	true	0.0546
joe_the_force_01	652	2024-02-06	true	0.0756
joe_the_force_01	653	2024-02-07	true	0.1173
joe_the_force_01	654	2024-02-08	true	0.1544

Showing 1 to 5 of 800 entries

Previous 1 2 3 4 5 ... 160 Next

KPI Chart (Remember to Click the 'Refresh' Button ↑↑↑)



KPI Data

CSV Excel Copy Print Show 5 entries

Search:

model	round	date_resolved	resolved	KPI
joe_the_aijoe_01	650	2024-02-02	true	0.0554
joe_the_aijoe_01	651	2024-02-03	true	0.1023
joe_the_aijoe_01	652	2024-02-06	true	0.1423
joe_the_aijoe_01	653	2024-02-07	true	0.2017
joe_the_aijoe_01	654	2024-02-08	true	0.2493

Showing 1 to 5 of 8,720 entries

Previous 1 2 3 4 5 ... 1,744 Next

KPI Chart (Remember to Click the 'Refresh' Button ↑↑↑)



Cumulative

No

Hide Pending

No

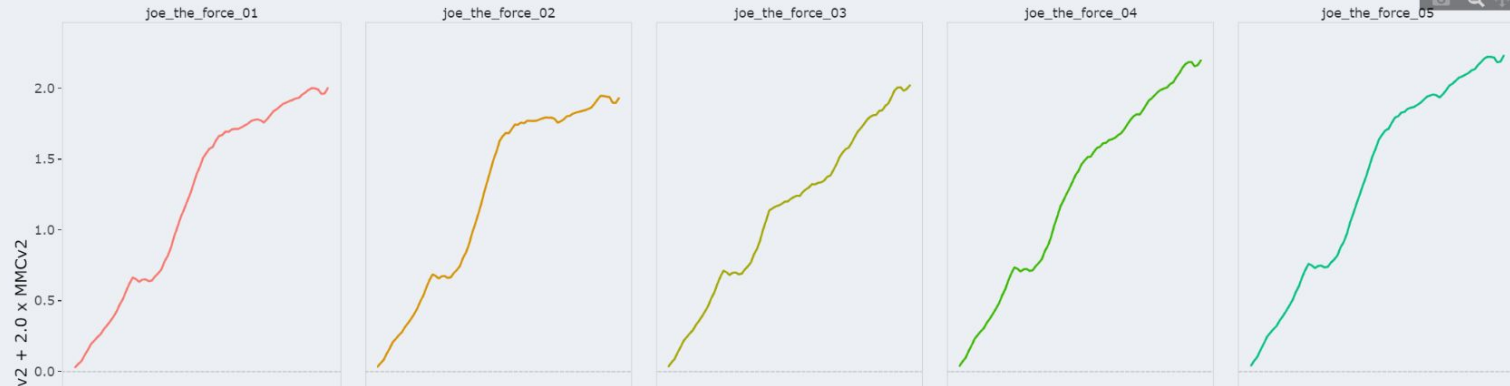
Facet

Yes



KPI Chart (Remember to Click the 'Refresh' Button ↑↑↑)

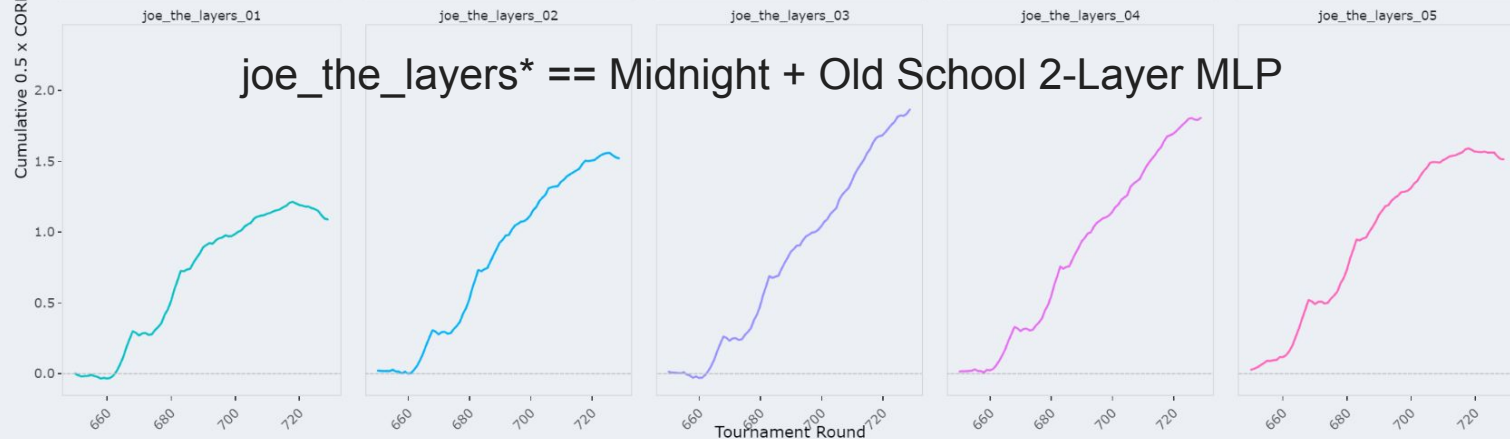
joe_the_force* == Midnight + XGB



model

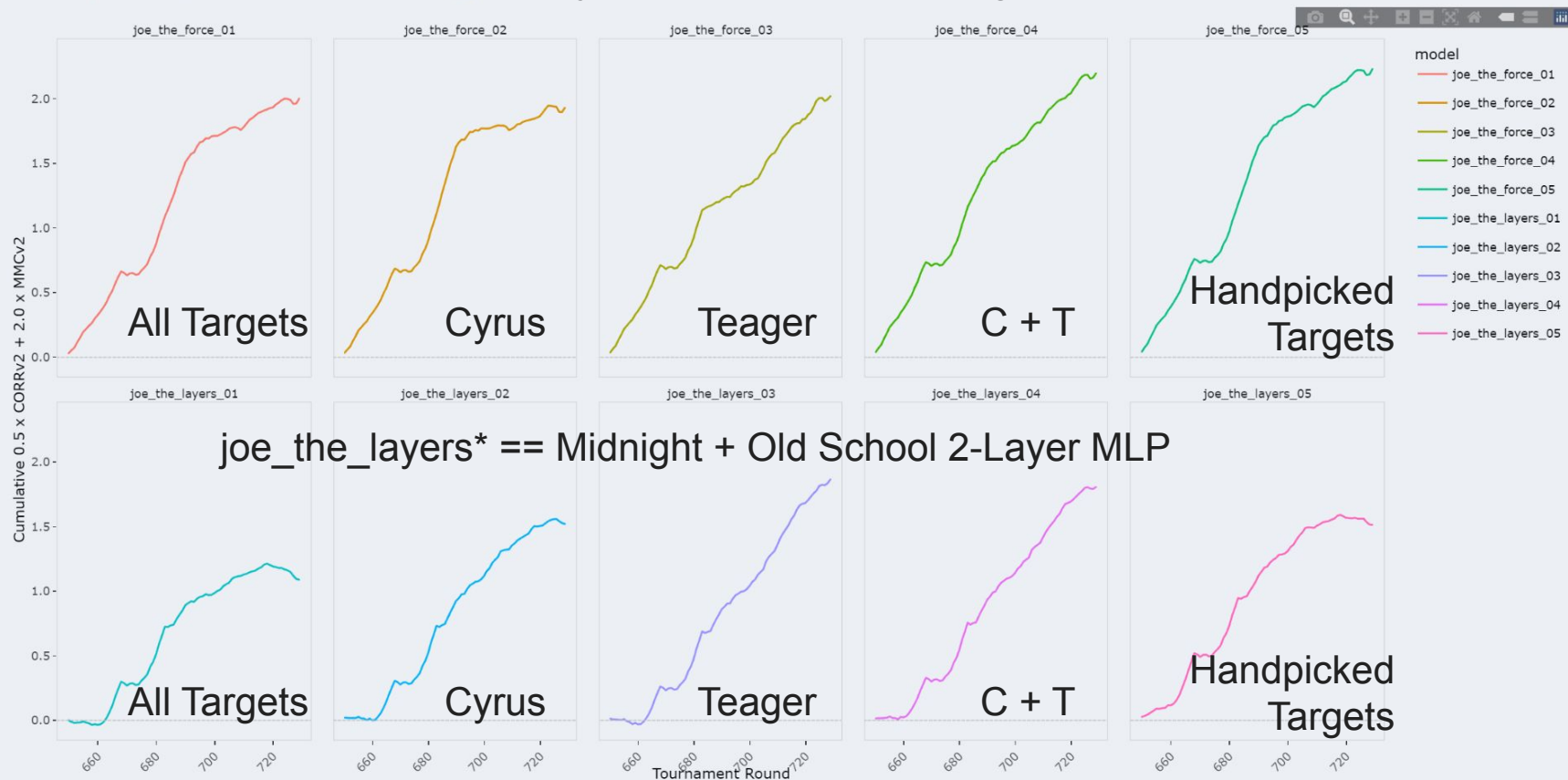
- joe_the_force_01
- joe_the_force_02
- joe_the_force_03
- joe_the_force_04
- joe_the_force_05
- joe_the_layers_01
- joe_the_layers_02
- joe_the_layers_03
- joe_the_layers_04
- joe_the_layers_05

joe_the_layers* == Midnight + Old School 2-Layer MLP



KPI Chart (Remember to Click the 'Refresh' Button ↑↑↑)

joe_the_force* == Midnight + XGB



[KPI \(All\)](#)[Payout \(Overview\)](#)[Payout Table \(Rounds\)](#)[Payout Table \(Models\)](#)[Payout Chart \(Rounds\)](#)[Payout Chart \(Models\)](#)

Payout Summary (Overview)

61

Staked Rounds (Resolved)

1348.37 NMR

Total Payout (Resolved)

22.10 NMR

Avg. Round Payout (Resolved)

0.49 %

Avg. Round ROR (Resolved)

19

Staked Rounds (Pending)

459.45 NMR

Total Payout (Pending)

24.18 NMR

Avg. Round Payout (Pending)

0.30 %

Avg. Round ROR (Pending)

80

Staked Rounds (All)

1807.82 NMR

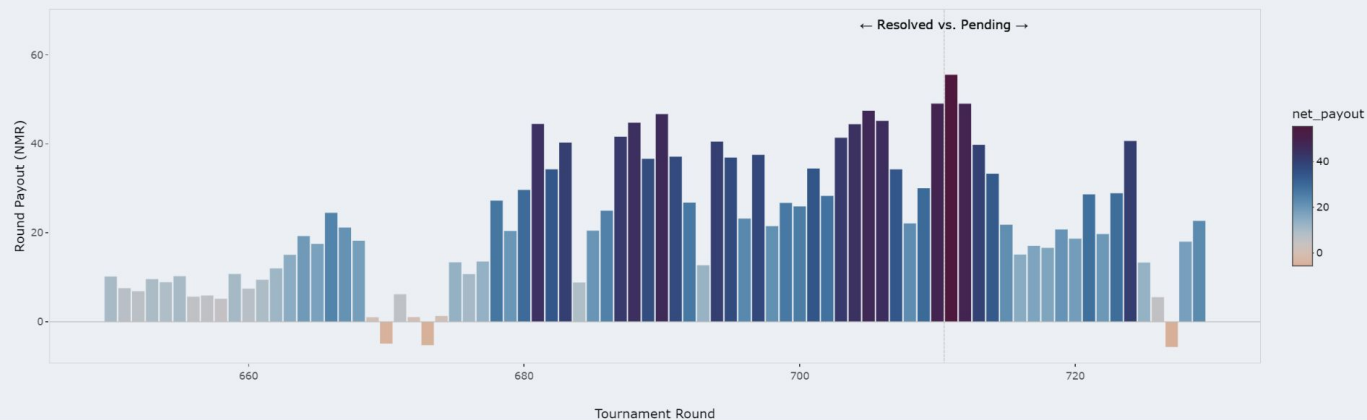
Total Payout (All)

22.60 NMR

Avg. Round Payout (All)

0.44 %

Avg. Round ROR (All)



Payout Summary (Individual Models)

CSV Excel Copy Print Show 100 entries

Search:

model	staked_rounds	net_payout	avg_payout	avg_rate_of_return_percent	sharpe_rate_of_return
joe_the_hatter_05	80	26.1747	0.3272	0.6194	2.5079
joe_the_wagie_05	80	26.0393	0.3255	0.5960	2.3970
joe_the_hatter_02	80	25.7454	0.3218	0.6022	2.2669
joe_the_rainbow_05	80	25.6528	0.3207	0.5958	2.2538
joe_the_game_05	80	24.4339	0.3054	0.5696	2.2092
joe_the_hatter_04	80	25.9834	0.3248	0.6140	2.2042
joe_the_game_02	80	24.6943	0.3087	0.5865	2.1409
joe_the_rainbow_01	80	21.4457	0.2681	0.4974	2.1174
joe_the_rainbow_04	80	24.1105	0.3014	0.5723	2.1074
joe_the_rainbow_02	80	24.5572	0.3070	0.5671	2.0953
joe_the_wagie_01	80	26.4462	0.3306	0.6188	2.0897
joe_the_game_04	80	23.3557	0.2919	0.5379	2.0607
joe_the_wagie_04	80	28.8750	0.3609	0.6709	2.0444
joe_the_rainbow_03	80	24.3115	0.3039	0.5592	1.9693
joe_the_wagie_03	80	27.1621	0.3395	0.6465	1.9654
joe_the_game_01	80	23.8940	0.2987	0.5388	1.9326
joe_the_game_03	80	23.4922	0.2937	0.5472	1.8788
joe_the_wagie_02	80	19.5638	0.2445	0.4981	1.8700
joe_the_hatter_03	80	25.8146	0.3227	0.5934	1.8489
joe_the_hatter_01	80	23.6118	0.2951	0.5631	1.8321
joe_the_hodler_04	80	25.6049	0.3201	0.5541	1.6641

NOTE: the charts may take a while to render if you have selected a lot of models.

Step 4: Adjust the Filter

Numerai Classic Tournament Rounds

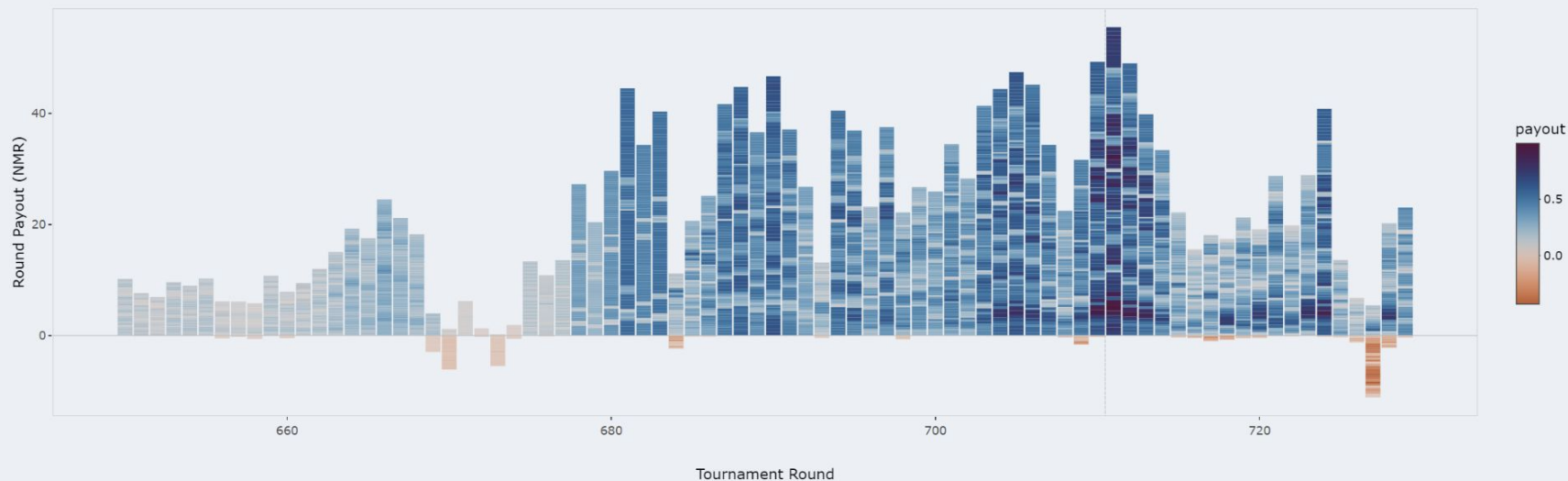


Step 5: Generate Summary

 Generate / Refresh

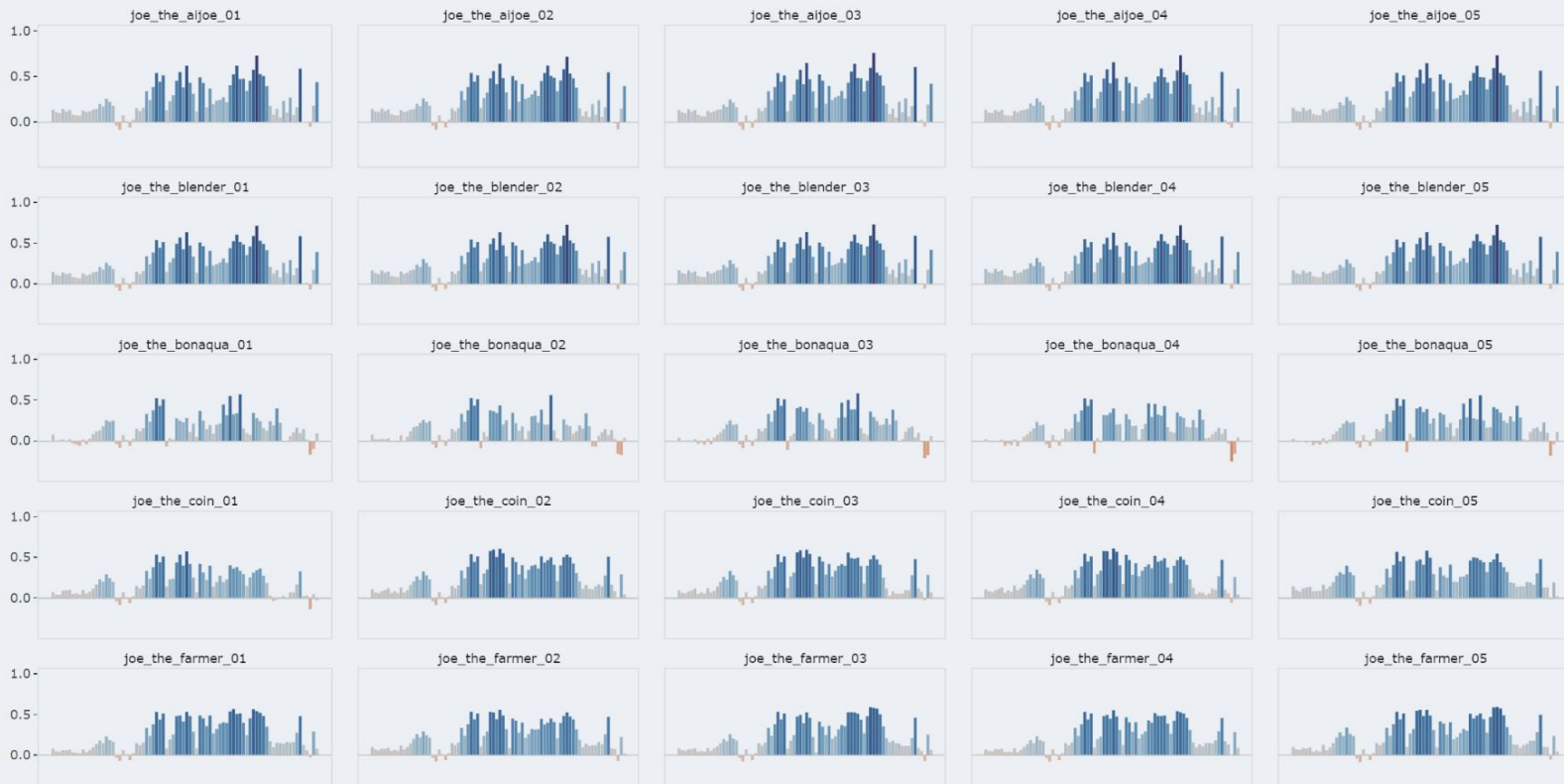
KPI (All) Payout (Overview) Payout Table (Rounds) **Payout Chart (Rounds)** Payout Chart (Models)

Payout Summary Chart (All Models - Stacked)



[KPI \(All\)](#)[Payout \(Overview\)](#)[Payout Table \(Rounds\)](#)[Payout Table \(Models\)](#)[Payout Chart \(Rounds\)](#)[Payout Chart \(Models\)](#)

Payout Summary Chart (Individual Models)



► Start Here

📊 Performance Summary

📄 Raw Data

👥 Community Events

🔍 About

Download Raw Data

Wanna run your own analysis? No problem.

Remember to select your model(s) first.

☁️ Download Raw Data CSV

▶ Start Here

📄 Performance Summary

📄 Raw Data

👤 Community Events

🔍 About

Numerai Coummunity

[About](#)[Materials](#)[Survey Results](#)[Memes](#)

Survey for Upcoming Events

Hello everyone!

We're reaching out to gather **anonymous data** on your location, which will greatly assist us in planning future Numerai community events.

Your support in providing this information would be much appreciated.

You can find the latest survey results below.

Here is the [Google Form](#). Please share it with your fellow Numeratis.

Thank you!

Summary

Copy Excel CSV Print Show 20 entries

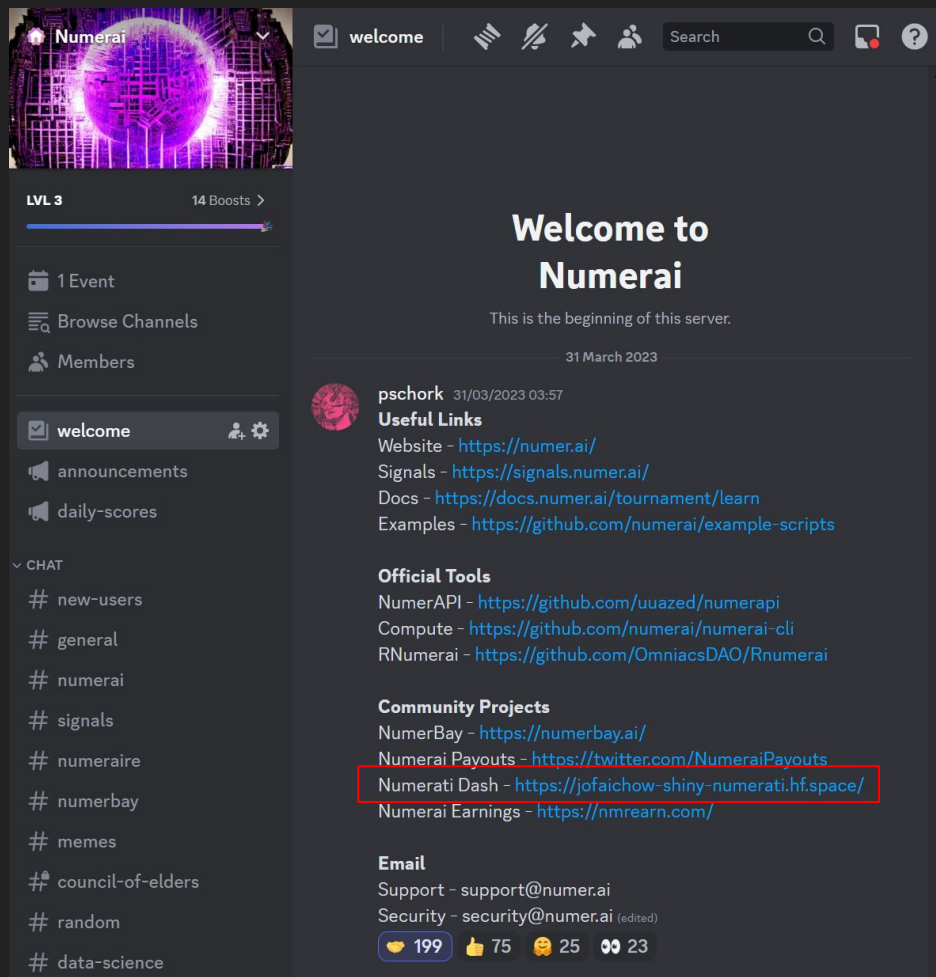
country	count
United States	27
Germany	16
Canada	11
Japan	10
United Kingdom	9
Spain	6
Australia	5
Switzerland	5
Denmark	4

Raw Data

Copy Excel CSV Print Show 20 entries

timestamp	country	comments
2023-11-20T17:04:54Z	Denmark	
2023-11-20T17:04:56Z	United States	
2023-11-20T17:05:10Z	Germany	
2023-11-20T17:05:38Z	Russia	
2023-11-20T17:06:03Z	Brazil	
2023-11-20T17:06:16Z	United Kingdom	
2023-11-20T17:06:28Z	Germany	
2023-11-20T17:06:35Z	United States	
2023-11-20T17:07:09Z	Spain	

Discord →
Welcome
<https://jofaichow-shiny-numerati.hf.space/>



The screenshot shows the Numerai Discord server interface. The top bar includes the server name 'Numerai', a search bar, and icons for settings, voice, and help. The left sidebar displays the server's level (LVL 3), boost count (14 Boosts), and a list of channels: # announcements, # daily-scores, # new-users, # general, # numerai, # signals, # numeraire, # numerbay, # memes, # council-of-elders, # random, and # data-science. The main content area features a 'Welcome to Numerai' message from the user 'pschork' dated 31/03/2023 03:57. The message includes 'Useful Links' (Website, Signals, Docs, Examples), 'Official Tools' (NumerAPI, Compute, RNumerai), 'Community Projects' (NumerBay, Numerai Payouts, Numerati Dash, Numerai Earnings), and 'Email' (Support, Security). The 'Numerati Dash' link is highlighted with a red box.

Welcome to Numerai
This is the beginning of this server.

31 March 2023

Useful Links
Website - <https://numerai.ai/>
Signals - <https://signals.numerai.ai/>
Docs - <https://docs.numerai.ai/tournament/learn>
Examples - <https://github.com/numerai/example-scripts>

Official Tools
NumerAPI - <https://github.com/uuzed/numerapi>
Compute - <https://github.com/numerai/numerai-cli>
RNumerai - <https://github.com/OmniacsDAO/Rnumerai>

Community Projects
NumerBay - <https://numerbay.ai/>
Numerai Payouts - <https://twitter.com/NumeraiPayouts>
Numerati Dash - <https://jofaichow-shiny-numerati.hf.space/>
Numerai Earnings - <https://nmrearn.com/>

Email
Support - support@numerai
Security - security@numerai (edited)

199 75 25 23

Feedback / Suggestion:
Forum post

or

Numerai Discord:
ia_ai_Joe



Introducing the Shiny Numerati Dashboard

Council of Elders



ia_ai

3 Nov 2023

Hey fam, as promised, I am writing this post about **Shiny Numerati**. It is a community dashboard for the main Numerai tournament. You can use it to track your model performance and payouts.

The main objectives of this forum post:

1. Onboarding new users with a simple quick-start guide
2. Collecting feedback from the community (bugs and feature requests)
3. Providing a roadmap (and polls to upvote feature requests from time to time)
4. Providing updates on new features

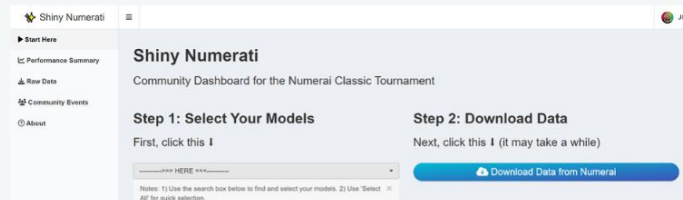
OK, let's get started!

1. Links to App:

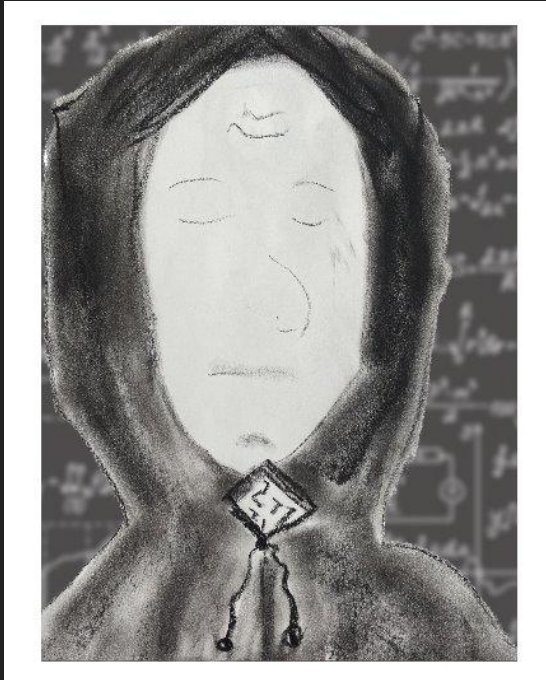
- App with HF nav bar: Shiny Numerati - a Hugging Face Space by jofaichow ⁶⁶
- App without HF nav bar: <https://jofaichow-shiny-numerati.hf.space/> ²⁹
- Source code (HF): jofaichow/shiny-numerati at main ⁴
- Source code (GitHub): GitHub - woobe/shiny-numerati ⁵

2. Quick-start Guide

Step 1: "Start Here" → Use the picker input widget to select your model(s) → Step 2: Download data from Numerai



Any Questions?



Contact

Jo-fai (Joe) Chow

Numerai Discord: ia_ai_Joe

Twitter/X: @matlabulous